# Manufacturing features

Higher · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ identify split lines, flashing, ejector marks, draft angles and injection points
- ☐ explain the purpose of bosses, location pins, ribs and webs

## I DO — Worked example

Read how a full-mark answer is built

**1 Explain the purpose of location pins, bosses and ribs/webs.** (4 marks)

**Full-mark answer** each ✓ is one mark

Location pins line the two halves up accurately during assembly ✓. Bosses provide a fixing point for the screws ✓ and strengthen the area around the fixing ✓. Ribs add rigidity to thin walls without making them thick, and help the plastic flow through the mould ✓.

**Why answers lose marks**

*"They help it fit together and make it strong."*

1 mark at most: not linked to each feature.

## WE DO — Guided practice

Use the starters and word bank to help you

**2 Describe how manufacturing features ensure accurate and efficient assembly of a product such as a hairdryer.** (2 marks)

**Sentence starters**

First, …

Next, … using a …

**Word bank**

exactly · together · screws · Ribs/webs · location · halves · internal · bosses

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Why are ribs added to the inside of a plastic casing?
☐ To make it heavier ☐ Add rigidity without thick walls and hold internal parts ☐ For grip ☐ To increase cost

**b)** Location pins mainly help with...
☐ Accurate assembly ☐ Corrosion resistance ☐ Aesthetics ☐ Recycling

**c)** What is the main purpose of a boss on an injection-moulded casing?
☐ Decoration ☐ To take a screw and strengthen the fixing point ☐ To let air escape ☐ To show the recycling code

**d)** Thin excess plastic along the edge of a moulded part is called...
☐ Draft ☐ Flashing ☐ Boss ☐ Rib

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Boss** | ____ | **A** Add strength and rigidity, help material flow, locate internal parts. |
| 2. **Location pin** | ____ | **B** Positions mating parts accurately during assembly. |
| 3. **Ribs/webs** | ____ | **C** Raised cylinder to receive a screw; adds strength at the fixing. |
| 4. **Flashing** | ____ | **D** Thin excess material at a mould's split line. |

### 5 Explain what is meant by ‘Split line’. (2 marks)

### 6 Explain what is meant by ‘Shear marks’. (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?